

DECODED EXPLAINED

28th August, 2026

Can India Build a Strategic Fuel System?

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India's proposed strategic-fuel programme would expand crude, LNG and LPG storage, but building reserves alone will not guarantee energy security; the system will work only if India can move the fuel through ships, pipelines and distribution networks, while having clear rules on ownership, financing, and replenishment

EXPLAINER

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The story so far:

West Asia-related disruptions have exposed a vulnerability in India's energy-security system. India can import large quantities of crude oil, LNG and LPG, but its ability to store, transport and release these fuels during a prolonged disruption differs sharply by fuel. India already operates underground strategic crude-oil storage and two underground LPG caverns, but has no operating underground natural-gas storage facility.

Against the background, the government is considering a decade-long strategic fuel programme that would add about 28 MT of crude-oil storage, 9 MT of LNG storage and 4 MT of LPG storage. Reuters reports that the proposed reserves are intended to provide nearly two months of crude and LNG demand and about six weeks of LPG demand. The programme remains under consideration and should therefore be treated as a proposed target, not committed capacity.

How much storage does India have today?

India's position varies considerably by fuel. For crude oil, Phase I of the Strategic Petroleum Reserve provides 5.3 MT of underground capacity, with actual storage of about 3.37 MT, or roughly 62-64% utilisation, across Visakhapatnam, Mangalore and Padur. A further 6.5 MT has been approved under Phase II at Chandikhol and Padur.

However, execution on the Chandikhol project remains protracted amid ongoing land acquisition requirements and finalisation of commercial PPP frameworks.

LPG has a much smaller underground footprint. The Visakhapatnam and Mangalore caverns together provide about 0.14 MT of capacity. A proposed 4 MT strategic reserve would therefore represent roughly a 30-fold increase in underground LPG capacity.

Natural gas presents the largest storage gap. India currently has no operational underground gas-storage facility. Its gas-security system instead depends on domestic production, LNG imports, import terminals, commercial inventories and pipelines.

These figures also highlight an important distinction between storage capacity and inventory. A facility has a physical capacity, but the fuel it holds can vary. Not all inventory is necessarily immediately accessible either. During a crisis, the critical measure is therefore not simply how many tonnes can be stored, but how much fuel is available and at what withdrawal rate it can reach consumers.

What does the proposed 9 MT of LNG storage mean?

LNG is particularly prone to confusion because India already has substantial LNG import and regasification infrastructure. A regasification terminal can receive LNG and convert it into natural gas, but its regasification capacity does not constitute strategic inventory.

LNG is stored as a cryogenic liquid at around -162°C, requiring specially insulated tanks and boil-off gas management. Underground gas storage is different: LNG is first regasified and the resulting natural gas is injected into a depleted reservoir or cavern.



India already operates underground strategic crude-oil storage and two underground LPG caverns, but has no operating underground natural-gas storage facility. GETTY IMAGES

The PNGRB ICF study provides a useful but narrowly defined stress test. Its 2030 scenario examined how much additional LNG infrastructure would be required to supply priority consumers for 20 days under stress conditions. It estimated a requirement of about 0.56-0.6 MT of LNG equivalent, achievable with roughly eight additional LNG tanks. The combined cost of the tanks and the associated strategic inventory was estimated at around \$1 billion. This was a scenario-specific estimate, not an assessment of India's overall strategic LNG requirement.

The proposed 9 MT strategic inventory is an order of magnitude larger. Against the ICF projection of roughly 58 MT of annual LNG imports, 9 MT represents about 56 days of imports – hence the description of approximately two months of LNG import cover. It does not mean two months of India's total gas consumption.

The difference is important. A 0.6 MT requirement addresses a 20-day priority-sector stress scenario; 9 MT represents a much broader strategic objective of buffering India against prolonged LNG-import disruption. The associated investment would therefore be substantially larger.

Does all the gas have to be stored as LNG?

No. India could eventually use a portfolio of surface LNG tanks and underground natural-gas storage.

Additional LNG tanks would hold inventory directly at import terminals. A separate proposal from the Ministry of Petroleum and Natural Gas would require LNG import terminals to maintain storage capacity 10% above their normal import requirement, with the additional capacity available to the government during supply or price disruptions.

For longer duration requirements, depleted oil and gas reservoirs, which account for 74% of global working gas volume, could provide much larger volumes of underground gas storage. Salt caverns offer different advantages: faster injection and withdrawal and the ability to cycle inventory more frequently, potentially making them useful for

shorter duration balancing.

But a depleted field cannot simply become a storage site because production has ended. Reservoir characteristics, cap-rock integrity, pressure behaviour, existing wells, cushion gas requirements and pipeline connectivity must all be assessed. India's sedimentary basins – including Krishna, Godavari, Cambay, Mumbai Offshore and Rajasthan – offer potential locations, but moving from geological potential to usable storage requires subsurface investigation, site selection, engineering, construction, testing, filling and pipeline integration.

Salt caverns face a similar qualification process. Rajasthan's salt-bearing formations have been investigated for solution-mined caverns, but suitability depends on depth, thickness, purity, geometry, groundwater and mechanical properties.

Can India scale its crude and LPG experience?

Crude is the most mature component. India already operates underground crude caverns and has developed the associated engineering capabilities. The U.S. Strategic Petroleum Reserve illustrates the scale possible with salt caverns, with authorised capacity of 714 million barrels across 60 caverns. But the U.S. system works as an integrated network of caverns, pipelines, marine terminals and refineries. Storage is therefore fundamentally an infrastructure system, not simply an underground space.

LPG is technically proven in India but presents a much larger scaling challenge. A proposed 4 MT reserve would require a substantial network of new caverns or other storage facilities, together with import terminals, pipelines, pumping systems and bottling infrastructure.

The Mangalore cavern illustrates the complexity. Underground construction requires geological and hydrogeological investigation, rock-mechanics analysis and groundwater management, particularly where work takes place alongside operating surface facilities. Scaling the system would require sustained capabilities in geophysics, geomechanics, tunnelling, reservoir engineering and project execution.

What happens if the fuel cannot reach India?

Because energy security relies heavily on maritime logistics, including vulnerable chokepoints such as the Strait of Hormuz and long-distance routes from the U.S., India is proactively diversifying both its import sources and shipping capabilities. To reduce reliance on foreign vessels, State-run oil refiners and the Shipping Corporation of India plan to invest \$5.5 to \$2 billion in a joint venture to acquire 59 ships. Simultaneously, Indian Oil is expanding sourcing through new 2027 agreements with Algeria and increased U.S. purchases, while exploring direct stakes in Very Large Gas Carriers to secure greater control over its supply chain.

What about pipelines?

Pipelines provide the final link between imported or stored fuel and inland consumers. PNGRB has recently authorised approximately 1,500 km of new LPG pipelines across six States, involving investment of around \$0.7 billion. These pipelines improve connections between supply sources and inland markets and reduce dependence on road movement, but they do not themselves add strategic storage. Their value lies in improving deliverability.

The same principle applies to natural gas. Underground storage is useful only if it can inject into and withdraw from the gas grid at the required rate. LNG tanks are useful only when regasification and downstream pipelines can move the gas. LPG caverns require connections to bottling and distribution systems.

Who pays for the strategic reserves?

The reported \$42 billion programme, which has yet to be confirmed by the government, combines infrastructure CAPEX with the cost of purchasing and maintaining strategic fuel inventories. Reuters reported that more than half could go towards storage infrastructure, with the balance used to purchase and fill the reserves. Financing would be a challenge and the government shut down reports that a cess would be charged to pay for the reserves.

The financial obligation does not end with construction. Billions of dollars of fuel would need to be purchased, stored and maintained. Stocks released during a crisis would then have to be replenished, potentially when commodity and freight prices are higher. The policy therefore needs to establish who owns and finances the inventory, minimum stock obligations, emergency release authority, and who bears replenishment and price risk.

A commercial-tank-strategic model could reduce the public burden, provided commercially used capacity remains available during emergencies.

Is storage enough to make India energy-secure?

Building 41 MT of combined storage is only part of the equation. A strategic reserve is effective only if India can seamlessly access fuel, transport it inland, and replenish stocks after a release.

Ultimately, energy security relies on an integrated framework: reserves buy time, shipping brings the next cargo, pipelines deliver fuel inland, and governance dictates emergency execution. The ultimate test of India's strategic fuel programme will be its ability to coordinate all these elements simultaneously when a crisis hits. (The author is an Energy and Emerging Technologies expert)

THE GIST

India's existing strategic storage offers sharply by fuel, with crude having the most developed system. LPG has limited underground footprint and natural gas no operational underground storage facility.

The proposed 28 MT of crude, 9 MT of LNG and 4 MT of LPG reserves would represent a major step forward, but the LNG requirement should not be confused with existing regasification capacity or narrower 20-day stress scenarios.

Geological suitability, infrastructure and project execution will determine how quickly storage can scale, while shipping and pipelines remain critical to ensuring fuel reaches consumers during prolonged disruptions.

Can India Build a Strategic Fuel System?

These are stockpiles of crude oil, maintained by the government to ensure energy security to be used during supply disruptions. These exist in addition to the commercial stocks held by oil companies.

Under Special Purpose Vehicle called Indian Strategic Petroleum Reserves Limited (ISPRL), established in 2004. Ministry of Petroleum and Natural Gas.

Phase-I SPR Facilities (5.33 MMT): Currently operational at three locations

Visakhapatnam (Andhra Pradesh)

Mangaluru (Karnataka)

Padur (Karnataka)

Phase-II (approved in 2021, with additional 6.5 MMT):

Chandikhol (Odisha)

Padur Phase-II

At full capacity, they can cover approximately 9.5 days of India's crude oil requirements.

The IEA (International Energy Agency) recommends that countries maintain oil stocks equivalent to at least 90 days of net oil imports. India's current total national storage capacity including commercial stocks with refiners stands at 74 days, still short of the 90-day IEA standard.

Why 543 Should Remain 543

Why 543 should remain 543

Days after both Houses of Parliament were adjourned *sine die* on August 13, 2026, the monsoon session has yet to be formally prorogued. The delay has prompted speculation that Parliament could be reconvened for another attempt at the constitutional amendment linked to delimitation and women's reservation. Whether or not that happens, it gives urgency to a question Parliament must confront: must delimitation necessarily mean an enlargement of the Lok Sabha?

Numbers alone cannot decide

At first glance, the issue looks like arithmetic. India's population has grown enormously since Lok Sabha seats were frozen half a century ago, so it seems natural to argue that representation must now expand correspondingly. But, constitutional design is not arithmetic alone. Article 81 requires representation broadly to correspond to population, "so far as practicable", while Article 82 provides for readjustment after every Census. Yet, Parliament has twice modified the mechanical application of this principle when another compelling national objective was at stake. In 1976, through the 42nd Constitutional Amendment, the inter-State allocation of seats was frozen on the basis of the 1971 Census so that States successful in family planning were not politically penalised for reducing population growth. The freeze was later extended by the 84th Amendment until the publication of figures from the first Census conducted after 2026. Parliament thus recognised that population matters, but it is not the only constitutional value at stake.

The government has reportedly suggested increasing every State's representation by roughly 50%, arguing that this would leave each with the same 'proportion' of Lok Sabha seats that it has today. On the face of it, that appears to answer the fear of the southern States of losing their relative position. But 'proportion' tells only half the story.

Suppose Uttar Pradesh rises from 80 seats to 120, while Tamil Nadu rises from 39 to about 59. Their proportional relationship remains unchanged. Yet, the absolute gap in voting strength rises from 41 Members of Parliament (MP) to around 61. Parliament does not vote in proportions; it votes in absolute numbers. Governments are formed by numbers, and confidence motions, legislation, and constitutional amendments are decided by numbers. A 50% pro-rata increase may preserve the ratio while widening the numerical gap.

The 1976 freeze therefore embodied a federal compact: States that succeeded in population stabilisation would not see their political voice diminished for doing what national policy encouraged. That understanding should not now be thoughtlessly discarded.

There is also a misconception that delimitation and enlargement are inseparable. They are not. India has already shown that constituency boundaries can be redrawn without changing the



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Delimitation, federal fairness and women's representation can all be achieved without enlarging the Lok Sabha

number of seats allocated to each State. The delimitation after the 2001 Census reorganised constituencies within States while maintaining the existing inter-State allocation.

That principle can be adopted again. The 2027 Census can provide the basis for redrawing constituencies within each State so that populations are more evenly distributed. Boundaries can reflect demographic movement and urbanisation. None of this requires increasing the Lok Sabha beyond 543 or disturbing the present federal balance.

A continuously expanding legislature is also not an inevitable consequence of population growth. The United States House of Representatives has remained at 435 voting members (there were some temporary exceptions) for more than a century despite enormous population growth. Switzerland's National Council has remained at 200 members since 1963. Going a step further, Hungary reduced its Parliament from 386 members to 199, while Italy reduced its Chamber of Deputies from 630 to 400. There is no democratic law requiring legislatures to grow whenever population grows.

Bigger House, weaker deliberation

There is another serious concern: what a larger House does to Parliament's own functioning. Parliamentary time does not expand with the number of MPs. Even with 543 members, many MPs hardly get opportunities to speak, ask questions or raise matters of public importance. A Lok Sabha enlarged by 50%, to more than 800 members, would reduce those opportunities further. It may increase representation numerically while weakening the quality of deliberation, one of Parliament's most important functions. Indian MPs already represent the largest average constituencies among the world's major democracies – several times the size of a U.S. congressional district or a U.K. constituency. Freezing the House at 543 while the population keeps growing means that burden only increases after delimitation, and the case for more representatives to improve accessibility is a real one. But it does not follow that the additional representatives must sit in Parliament. Most citizens approach elected representatives over roads, schools, hospitals, local services and State administration – matters substantially within the State sphere. If population growth has made constituencies unwieldy, strengthening representation at the State level is, therefore, a more sensible solution. More MLAs and smaller Assembly constituencies can bring representatives closer to citizens without continuously enlarging the national legislature.

That expansion faces none of the federal-arithmetic constraint that makes Lok Sabha expansion fraught. A larger Assembly shifts no balance between States – each answers only to its own population. Vidhan Sabhas can therefore grow to meet accessibility needs without disturbing the inter-State balance in Parliament that the 1976 freeze was designed to protect. Then comes women's reservation.

One-third reservation for women is now being connected with delimitation, and enlargement of the Lok Sabha is presented partly as a way of accommodating women without disturbing existing political interests.

But women's representation should not become a constitutional justification for enlarging the House. One-third of the present Lok Sabha is approximately 181 seats. Women can receive their constitutionally promised representation within the existing House of 543. The purpose of reservation is to alter who occupies parliamentary seats, not necessarily to create hundreds of additional ones.

A 50% expansion could produce a curious result: the number of women MPs would rise, which is desirable, but so would the numerical dominance of the already larger States. A reform intended to correct one representational imbalance could deepen another. Women's representation and federal fairness are both legitimate constitutional objectives. India need not choose between them.

Preserve 543, undertake reforms

The solution is simple: retain the Lok Sabha at 543; preserve the present inter-State distribution of seats; undertake delimitation within States on updated population figures; implement women's reservation within that structure; and expand the Vidhan Sabhas to absorb the accessibility pressure that population growth has created.

For half a century, States have pursued demographic policies knowing that successful population stabilisation would not reduce their parliamentary influence. Southern States in particular invested heavily in education, health care and family welfare and achieved fertility levels that national policy itself encouraged. To reward that success with a larger political advantage for States where population grew faster would invert the incentive the Union deliberately created.

The question, therefore, is not whether population should matter. Of course it should. The question is whether it should now mechanically rearrange the federal political balance that India consciously preserved for five decades. Twice before, Parliament amended the constitutional framework to preserve the existing inter-State allocation despite population changes. There is no reason it cannot do so a third time. The uncertainty over whether Parliament may be recalled makes this more than an academic debate. Decisions taken now could shape the federal balance of the Republic for decades.

Delimitation is necessary. Women's representation is overdue. Updated constituency boundaries are desirable. But none of these objectives requires an 800-plus-member Lok Sabha.

Sometimes constitutional wisdom lies not in making institutions larger, but in preserving the balance that has allowed them to endure; 543 has served India for more than half a century. There is a strong constitutional, federal and democratic case for letting this number remain 543.

Why 543 Should Remain 543

- **The Trigger**

- Days after both Houses were adjourned sine die on August 13, 2026, the monsoon session has yet to be formally prorogued, fuelling speculation Parliament could reconvene for another attempt at the constitutional amendment linked to delimitation and women's reservation.
- This raises a core question: must delimitation necessarily mean enlarging the Lok Sabha?

- **Why Numbers Alone Cannot Decide This**

- Article 81 requires representation to broadly match population 'so far as practicable', while Article 82 provides for readjustment after every Census — yet Parliament has twice modified this when another national objective was at stake, freezing inter-State allocation at 1971 Census levels via the 42nd (1976) and 84th Amendments.
- **The government has reportedly suggested a roughly 50% increase for every State, preserving each State's 'proportion' of seats but proportion tells only half the story, since Parliament votes in absolute numbers, not proportions.**

- **Delimitation Doesn't Require Enlargement**

- India has already shown boundaries can be redrawn without changing seat allocation — the 2001 Census delimitation reorganised constituencies within States while preserving the existing inter-State balance; the same could apply again after the 2027 Census.
- **A continuously expanding legislature isn't inevitable: the US House has stayed at 435 for over a century, Switzerland's National Council at 200 since 1963, while Hungary and Italy have actually shrunk their legislatures.**

BRICS Is Now an Engine of Growth for the World

BRICS is now an engine of growth for the world



QIN JIE

We are both at a critical stage of national development. Cooperation brings common prosperity, while confrontation can lead to a lose-lose situation

THE IDEA of "BRIC" was first mooted by the former chief economist of Goldman Sachs, Jim O'Neill, in 2001. The four-nation group had its first high-level ministerial meeting in 2006, and held its inaugural summit in 2009. South Africa officially joined the group in December 2010, and BRICS emerged.

In Chinese, BRICS is referred to as *jīnzhuan* (golden brick), which emphasises its significance. Today, BRICS+ has expanded to include 11 member countries and 10 partner countries. BRICS and BRICS+ now account for 45 per cent and 55 per cent of the world's population respectively and are rapidly closing in on 35 per cent to 45 per cent of global GDP (by purchasing power parity; even in nominal terms, BRICS+ economies collectively surpass \$30 trillion in GDP). This collective voice cannot be neglected. BRICS+ is emerging as the world's most dynamic economic alliance, and according to international agencies like the IMF, it is to grow nearly three times faster than the G7 nations by 2028. As ancient civilisations that led the world for centuries, BRICS countries are on their way to revival. More countries like Argentina, Algeria, and Türkiye have expressed interest in formally joining the group. These facts show that BRICS provides a special platform for dialogue and cooperation.

In less than a month, the 2026 BRICS Summit will be held in New Delhi. As an important platform for cooperation, BRICS+ has to expand its agenda to respond to changing global realities while remaining focused on economic growth and people-centric development. Following India's 2026 agenda, the four priorities are resilience, innovation, cooperation and sustainability, which will advance through three pillars of the group: Political and security cooperation, economic and financial partnership, and cultural and people-to-people exchanges.

China has pledged firm support for India's role in the summit. Given their size, whether in population, economy, or markets, cooperation is the best choice for our two countries. For instance, our economies are the first and third largest in terms of purchasing power parity (PPP), and their combined population is about 35 per cent of the world's total. We are both at a critical stage of national development. Cooperation brings common prosperity, while confrontation and antagonism can lead to a lose-lose situation. Recently, multiple direct flights between China and India have resumed. Exchanges and cooperation between sister cities, universities, and businesses have been increasing.

Of course, China-India solidarity benefits not only our two countries, but the region and the world. In the near future, China and India will remain strong engines of global economic growth. Our combined GDP will climb from the current 28.4 per cent using PPP terms, and roughly 20 per cent in nominal terms, of the world's total. The Five Principles of Peaceful Coexistence (Panchsheel) advocated by our two countries have become the fundamental principles of international law. Under our leaders' guidance, China and India can build more synergy and achieve mutual success. In the long run, more and more youth from both countries can become bridges between China and India.

The writer is consul general of the People's Republic of China in Mumbai

BRICS Is Now an Engine of Growth for the World

- **BRICS is a grouping of Brazil, Russia, India, China, and South Africa**
- **New members are Iran, UAE, Ethiopia, Egypt and Indonesia.**

The grouping consists of:

- **42% of the world's population**
- **30% of the world's territory**
- **23% of global GDP**
- **18% of world trade.**

Thank You